Tax Design with Labor Market Frictions

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Taxing with Frictions

"`Since the labor market is the main source of income for most individuals, the amount of frictional wage inequality should be informative for anyone interested in... the provision of social insurance`."

- Hornstein-Krusell-Violante, AER, 2011

- Worker's pay depends on surplus produced and share of surplus kept.
- Low pay because of *low talent* or because matched with an *extractive firm*.
- *How should policy be designed in face of ex ante dispersion in worker talent and ex post risk of moving up, falling off or getting stuck on the job ladder?*
- Do frictions strengthen the case for redistribution???

Taxing with Frictions: Framework

- Traditional normative PF models abstract from frictions.
  - Mirrlees 1971; Saez 2001
- Traditional search models abstract from income taxes.
  - Burdett-Mortensen (1998)
- Our paper combines elements of both:
  1. Private variation in worker talent.
  2. Intensive effort margin.
  3. On and off the job search $\Rightarrow$ job ladders.
  4. Job creation and matching.
  5. Taxes and benefits.

Taxing with Frictions: Preview

- Higher marginal income taxes:
  1. Deter effort (as in frictionless model).
  2. Deter participation (as in frictionless model with participation).
  3. Impact job ladders. (**Profit Channel**).
- Higher taxes force low paying firms to pay more to more to attract workers.
- Competition amongst firms causes higher paying firms to raise worker incomes too. Profits are squeezed.
- Raises income tax revenues; Lowers profit tax revenues; Deters vacancy creation.
- Redistributes within and across talent markets.

Taxing with Frictions: Preview of Results

- Theory: `New optimal tax formulas`.
  - Show how labor market frictions modify existing formulas.
- Quantitative: `Model calibrated to U.S. economy`.
  - Several countervailing forces. Overall, frictions a force for (moderately) lower optimal marginal taxes.

Literature

- Taxation with Exogenous Wages
  - Mirrlees (1971); Diamond (1998); Saez (2001).
- Taxation with Endogenous Wages
  - Stiglitz (1982); Rothschild-Scheuer (2013, 2014); Slavik and Yazici (2014); Ales-Kurnaz-Sleet (2015); Scheuer-Werning (2015,2016); Ales-Sleet (2016), Sachs-Tsyvinski-Werquin (2016), Stantcheva (2014).
- Taxing with Search Frictions
  - Boone-Bovenberg (2002, 2006); Hungerbühler et al (2006); Lehmann et al (2012), Golosov-Maziero-Menzio (2013); Bagger et al (2017).
- Search Frictions
  - Burdett-Mortensen (1998); Mortensen (2000); Bontemps et al. (2000); Postel-Vinay and Robin (2002); Shimer (2012).

Environment

Model: Basic Elements

- Time continuous.
- Attention restricted to steady state equilibria and time invariant policy.
- Workers and firms trade effort for income in frictional labor markets segmented by talent.

Talent, Technology, Markets

- **Talent:** $\theta \in [\underline{\theta}, \overline{\theta})$ drawn at beginning of life.
  - Talent distribution K; density k.
  - Talent converts effort into effective labor: $z = \theta e$.
- **Technology:** Firms own technology.
  - $y = z$.
- **Markets:** Firms observe worker talent θ.
  - Effectively, labor market partitioned into (frictional) talent markets.

Decisions

- **Firm decisions:**
  - θ talent market, vacancies v and contract.
  - Efficient contract summarized by "take it or leave it" job price q.
- **Worker decisions:**
  - Accept/reject contract if encountered.
  - If worker accepts, then chooses effort and consumption given taxes, job price.

Worker Preferences

✲ **Preferences** defined over *consumption-effort allocations*:

$$U(c,e)$$

- $U(\cdot,e)$ strictly concave and increasing in c.
- $U(c,\cdot)$ strictly concave, decreasing in e.

Taxes and Benefits

- **Taxes on Workers:**
  - **Employed:** pay tax $T[x]$ on income x.
    - *Affine*: $T[x] = -L + \tau x$.
    - *Nonlinear*: T general function.
  - **Unemployed:** receive b.
- **Taxes on Firms:**
  - profits (optimally) taxed at 100%.
- **Policy notation:** $\mathcal{P} = (b, T)$.

Budgets and Worker Choices

- **Earnings:** θ-worker exerting e at firm with job price q earns:

$$x = \theta e - q.$$

- **Budget:** Faces budget constraint:

$$c = L + (1-\tau)x = L + (1-\tau)\{\theta e - q\}$$

- **Choice:** Once matched workers choose e to solve:

$$\Phi(q, \theta; \mathcal{P}) := \max_{e} U\left(L + (1-\tau)\{\theta e - q\}, e\right).$$

Getting out of Unemployment

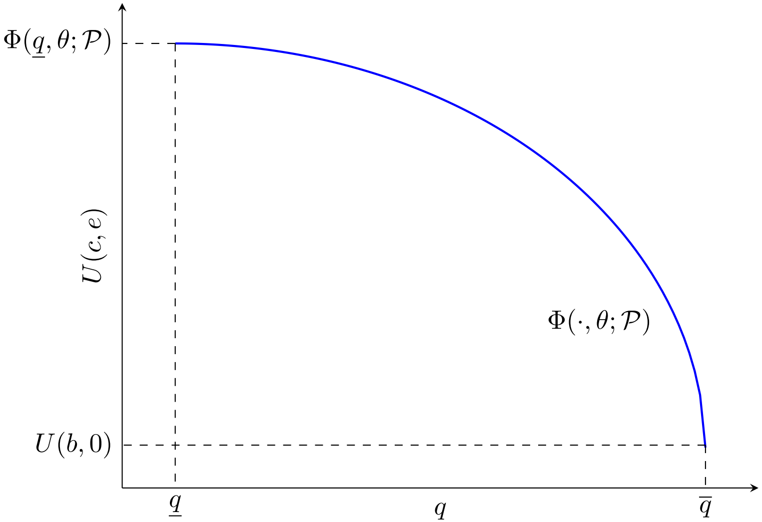
* **Unemployed θ worker**:

  1. meets firm with probability $\lambda(\theta;\mathcal{P})$.
  2. conditional on meeting, draws job price q' from $F[\cdot|\theta;\mathcal{P}]$.
  3. accepts offer and takes job if $q' \leq \overline{q}(\theta;\mathcal{P})$, where:

$$\Phi(\overline{q}(\theta;p),\theta) = \max_e U\left(L + (1-\tau)\{\theta e - \overline{q}(\theta;\mathcal{P})\}, e\right) = U(b,0).$$

* **Maximal q**:
  - $\overline{q}(\theta;\mathcal{P})$ is maximum job price firm can ask in market θ.
  - $\overline{q}(\theta;\mathcal{P})$ is decreasing in τ.

Utilities and Job Prices



Climbing the Ladder

- **Employed θ worker at firm with job price q:**

  1. meets new firm with prob. $\lambda(\theta;\mathcal{P})$,
  2. conditional on meeting, draws job price q' from $F[\cdot|\theta;\mathcal{P}]$.
  3. accepts and moves if $q' < q$.
  4. if accepts and moves and solves:

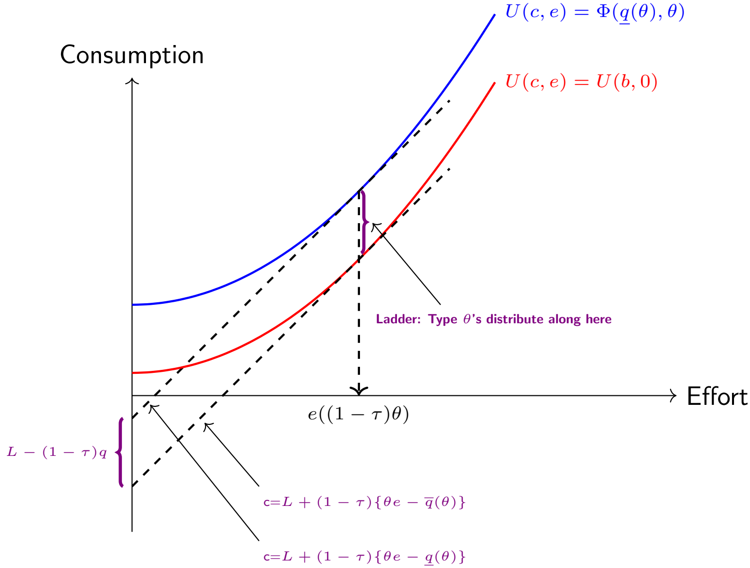
$$\Phi(q,\theta;\mathcal{P}) := \max_{e} U\left(L + (1-\tau)\{\theta e - q\}, e\right).$$

☞ Moving **up** ladder means moving to **lower** job price, **higher** income.

Falling Off the Ladder

- Employed worker jobs destroyed with prob. δ.
- Such workers enter unemployment pool.

Consumption and Effort and Taxes



Low and High θ

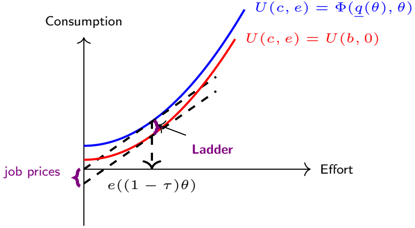


Figure: Worker choice at low θ and high θ'.

- Frictions enhance income dispersion in talent markets, but suppress it across.

Activity Threshold $\tilde{\theta}$

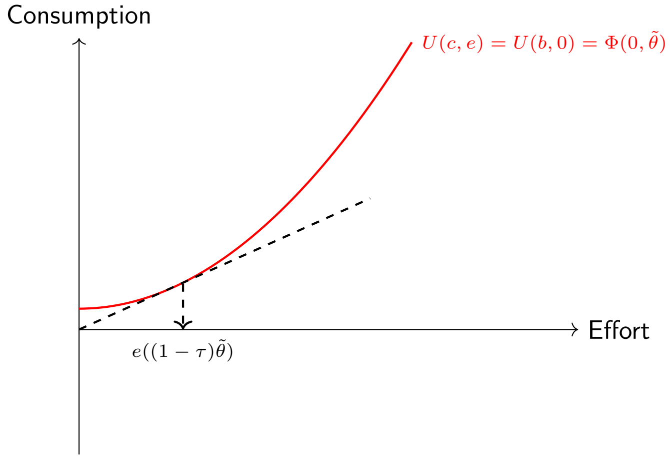


Figure: $\tilde{\theta}$ such that worker indifferent between working and not when $q = 0$.

Firm Problem

- **Firm choice**: Firms pick vacancies v and job prices q in *active* talent markets $\theta \in (\tilde{\theta}(\mathcal{P}), \overline{\theta})$ to max. steady state flow profit:

$$\max_{v,q} \quad \underbrace{N(q;\theta,\mathcal{P})}_{\substack{\text{No. of workers} \\ \text{per vacancy} \\ \text{in steady state}}} q \cdot v - \underbrace{\kappa(v;\theta)}_{\text{Vacancy cost}} .$$

- **Tradeoff**: Extractive (high q) & small (low N) vs. generous (small q) & large (high N).

- **Job price dispersion**: Firms distribute over a set of q's over which indifferent.

- **Basic Burdett-Mortensen (bBM)**: $\kappa(v;\theta) = \begin{cases} 0 & v \leq 1 \\ \infty & v > 1 \end{cases}.$

Job Price Setting

- Steady state conditions + firm problem $\implies$ eqm conditions for F (offer distribn of job prices) and G (worker distribn over job prices).

- Inverting latter, job price is (as function of G-quantile i):

$$q(i,\theta;\mathcal{P}) = \left(\frac{1}{1+(\lambda(\theta;\mathcal{P})/\delta)(1-i)}\right)^2 \overline{q}(\theta;\mathcal{P})$$

with:

$$\max_e U\left(L+(1-\tau)\{\theta e-\overline{q}(\theta;\mathcal{P})\},e\right) = U(b,0).$$

Matching

* **Standard matching technology**:

$$m(v(\theta;\mathcal{P}), k(\theta);\mathcal{P}) = \chi v(\theta;\mathcal{P})^{\alpha} k(\theta)^{1-\alpha},$$

where $v(\theta;\mathcal{P}) =$ vacancies created in talent market θ by firms.

* **Equilibrium job contact rates for workers**:

$$\lambda(\theta;\mathcal{P}) := \chi \left(\frac{v(\theta;\mathcal{P})}{k(\theta)}\right)^{\alpha}.$$

  - **For bBM model**, $\lambda(\theta;\mathcal{P}) = \lambda(\theta)$.

Impact of $\uparrow \tau$ on Vacancies and Employment Rates

- Derive v first order condition (at $\overline{q}(\theta)$) form:

$$\max_{v} \quad N(\overline{q}(\theta);\theta,\mathcal{P})\overline{q}(\theta)\cdot v - \kappa(v;\theta).$$

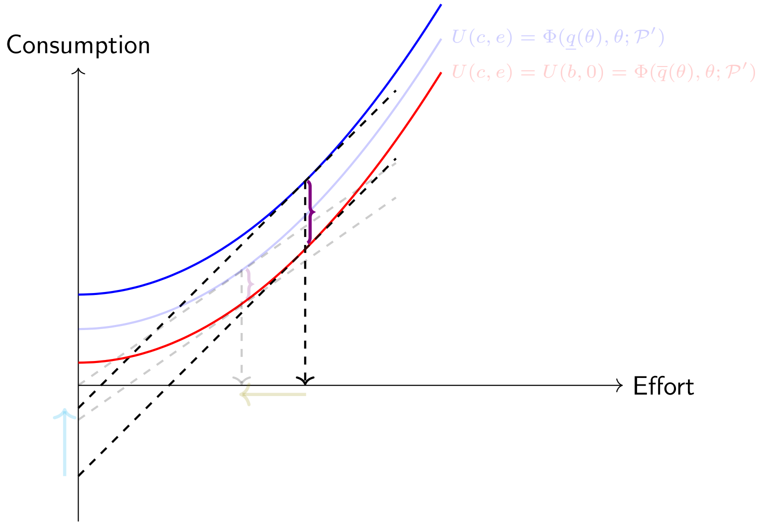
- Fall in revenues discourages vacancy posting:

$$\tau \uparrow, \qquad \underbrace{(\delta + D(\theta)v(\theta)^{\alpha})^{2}v(\theta)^{1-\alpha}}_{\downarrow} = E(\theta)\overbrace{\overline{q}(\theta)}^{\downarrow}.$$

- Lowers job contact rates and employment rates:

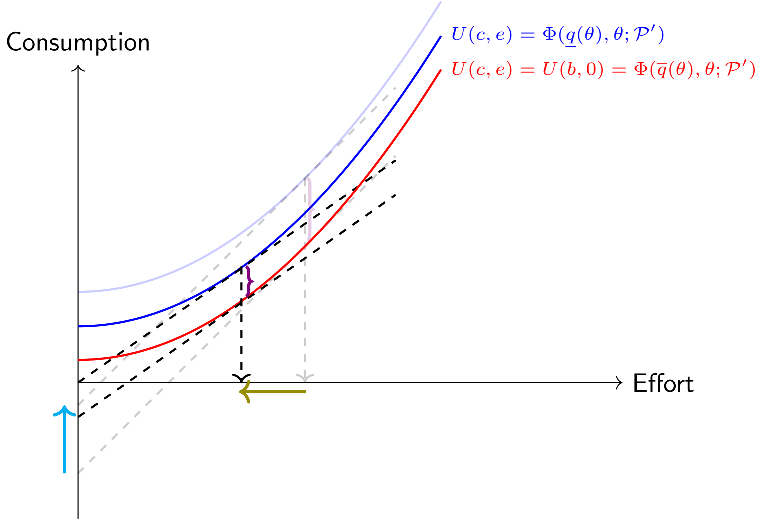
$$\tau \uparrow, \qquad \overbrace{\lambda(\theta)}^{\downarrow} := \chi\left(\frac{\overbrace{v(\theta)}^{\downarrow}}{k(\theta)}\right)^{\alpha} \qquad \overbrace{1-u(\theta)}^{\downarrow} = \frac{\overbrace{\lambda(\theta)}^{\downarrow}}{\delta+\lambda(\theta)}.$$

Tax Incidence



- A higher tax rate deters effort and compresses job prices.
- Firms at base of ladder must reduce job prices to attract workers.

Tax Incidence



- A higher tax rate deters effort and compresses job prices.
- Firms at base of ladder must reduce job prices to attract workers.

Incidence of $\uparrow \tau$ on Job Prices

- Effect of tax increase on job prices:

$$q(i,\theta) = \left(\frac{1}{1 + \overbrace{\frac{\lambda(\theta)}{\delta}}^{\downarrow}(1-i)} \right)^2 \overbrace{\bar{q}(\theta)}^{\downarrow}.$$

☞ **b-BM**:

$$\mathcal{E}_{q,(1-\tau)} = \frac{\underline{x}(\theta)}{\bar{q}(\theta)} > 0.$$

☞ **BM**:

$$\mathcal{E}_{q,(1-\tau)} = \left\{ 1 - 2\left(\frac{(\lambda(\theta)/\delta(\theta))(1-i)}{1 + (\lambda(\theta)/\delta(\theta))(1-i)} \right) \mathcal{E}_{\lambda/\delta,\bar{q}}(\theta) \right\} \frac{\underline{x}(\theta)}{\bar{q}(\theta)} > 0.$$

Job price Implications of tax rate rise

1. Job prices fall all along the job ladder: $-\frac{\partial q(i,\theta)}{\partial \tau} > 0$.
2. Job prices fall greatest at bottom of ladder: $-\frac{\partial q(i,\theta)}{\partial \tau}$
3. "`Price cutting`" and "`competition dampening`" channels.
4. Distribution of job price cuts across θ markets more complicated.
   - In b-BM mode, greatest at high θ.

Optimal Redistributive Taxation

Policymaker's Problem

- Utiliarian policymaker maximizes expected payoff subject to budget constraint:

$$\max_{\mathcal{P}} \int_{\underline{\theta}}^{\overline{\theta}} \left\{ u(\theta)U(b,0) + (1-u(\theta)) \int_0^1 \Phi(q(i,\theta),\theta)di \right\} K[d\theta]$$

subject to

$$-\mathcal{G} - \int_{\underline{\theta}}^{\overline{\theta}} [bu(\theta) + L(1-u(\theta))]K[d\theta]$$
$$+ \int_{\underline{\theta}}^{\overline{\theta}} (1-u(\theta)) \int_0^1 \tau x(q(i,\theta),\theta)diK[d\theta] + \Pi \geq 0. \qquad \text{(GBC)}$$

Optimal Tax Formula

$$\underbrace{-E\left[\left\{\frac{U_c}{\Lambda}+\frac{\tau}{1-\tau}\eta-1\right\}\left\{x+(1-\tau)\frac{\partial q}{\partial \tau}\right\}\right]}_{\text{Redistributive benefit}}\ \underbrace{-\frac{\tau}{1-\tau}E[x\mathcal{E}^{c}_{x,1-\tau}]}_{\text{Intensive margin revenue loss}}$$

$$\underbrace{-\frac{1}{1-\tau}E\left[\left(\left\{\frac{\Delta U}{\Lambda}+\tau x+b-L\right\}\mathcal{E}_{1-u,v}+\left\{\mathcal{E}_{1-u,v}-\frac{vN(q)}{1-u}\right\}q\right)\mathcal{E}_{v,1-\tau}\right]}_{\text{Social value of vacancy (employment) suppression}}=0.$$

- Red terms present in standard model; blue terms appear due to frictions.
- Redistributive benefit augmented by GE effect, $(1-\tau)\frac{\partial q}{\partial \tau}$.
- Last term calls for lower tax if there's too little vacancy posting to begin with.

Marginal Social Value of a Vacancy

$$\underbrace{-\frac{1}{1-\tau}E\left[\left\{\frac{\Delta U}{\Lambda}+\tau x+b-L\right\}\mathcal{E}_{1-u,v}\mathcal{E}_{v,1-\tau}\right]}_{\text{Worker utility and tax benefits of vacancy}}+\underbrace{E\left[\left\{\mathcal{E}_{1-u,v}-\frac{vN(q)}{1-u}\right\}q\mathcal{E}_{v,1-\tau}\right]}_{\text{Congestion and poaching costs}}$$

- **First term**: Firms do not internalize utility and tax revenue benefits of a vacancy.
- **Second term**: Firms do not internalize congestion and poaching costs of a vacancy.

Optimal Tax Formula

$$\underbrace{-E\left[\left\{\frac{U_c}{\Lambda}+\frac{\tau}{1-\tau}\eta-1\right\}\left\{x+(1-\tau)\frac{\partial q}{\partial \tau}\right\}\right]}_{\text{Redistributive benefit}} \quad \underbrace{-\frac{\tau}{1-\tau}E[x\mathcal{E}^c_{x,1-\tau}]}_{\text{Intensive margin revenue loss}}$$

$$\underbrace{-\frac{1}{1-\tau}E\left[\left(\left\{\frac{\Delta U}{\Lambda}+\tau x+b-L\right\}\mathcal{E}_{1-u,v}+\left\{\mathcal{E}_{1-u,v}-\frac{vN(q)}{1-u}\right\}q\right)\mathcal{E}_{v,1-\tau}\right]}_{\text{Social value of vacancy (employment) suppression}}=0.$$

- OTEs identify marginal benefits/costs of tax variation at optimum.
- But these are all endogenous variables.

Calibration

- **Worker preferences**

$$U(c,e) = \frac{1}{1-\sigma}\left(c - \frac{1}{1+\gamma}e^{1+\gamma}\right)^{1-\sigma}, \quad \sigma = 2, \gamma = 1.$$

- **Labor market**
  - $\delta = 0.03$ (monthly, Shimer, 2012).
  - $\alpha = 0.28$ (Shimer, 2005); $\chi = 1$ (normalization).

- **US government tax policy**
  - Affine approximation: $T[x] = -4230 + 0.338\, x$.

- **Internally calibrated**
  - $\kappa(v,\theta) = \kappa(\theta)v$ and $k(\theta)$.

Recovering $\kappa(\theta)$ and $k(\theta)$

- Recovering talent distribution in frictionless model simple:

$$h(x) = k(\theta(x))\frac{\partial \theta}{\partial x} \implies k(\theta) = h(x(\theta))\frac{\partial x}{\partial \theta}$$

- With frictions, map from talent to income density of employed h :

$$h(x) = \int_{\tilde{\theta}}^{\overline{\theta}} P(x|\theta)\frac{\lambda(\theta)}{\delta + \lambda(\theta)}\frac{k(\theta)}{\mathcal{N}}d\theta, \qquad \text{(E)}$$

$P(x|\theta) =$ kernel relating talent to income, $\mathcal{N}$ employment rate among $\theta \geq \tilde{\theta}$.

1. Closed form expression for $P(x|\theta)$.
2. Given $\lambda(\theta)$, (E) is a Fredholm equation of the first kind and can be inverted.
3. Could then recover $k(\theta)$ from empirical earnings distribution $h(x)$.
4. But don't know $\lambda(\theta)$.

Recovering $\kappa(\theta)$ and $k(\theta)$

✲ Map from talent to last-earned income density of currently *unemployed* l is:

$$l(x) = \int_{\tilde{\theta}}^{\overline{\theta}} P(x|\theta) \frac{\delta}{\delta + \lambda(\theta)} \frac{k(\theta)}{\mathcal{U}} d\theta, \tag{U}$$

$\mathcal{U} =$ fraction of agents with talents in excess of $\tilde{\theta}$ who are unemployed.

- Key observation: P same for employed & unemployed (since δ independent of place on job ladder).
- Invert: (E)-(U). Get estimates of $k(\theta)$, $\lambda(\theta)$ (wh. gives $\kappa(\theta)$) from $h(x)$, $l(x)$.

Optimal Affine Policy

	Frictional	Frictionless
τ	31.7%	35.7%
L	2148	5220
b	9060	8580

L, b: *annual 2017 US $ amounts.*

- Frictions call for (moderately) lower optimal marginal income tax.

Optimal Affine Policy

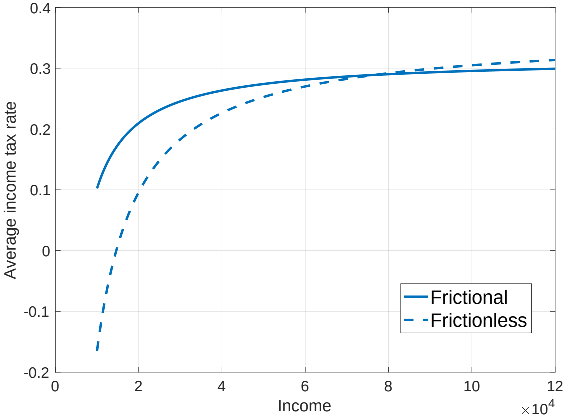


Figure: Average income tax rates $\left(\frac{\tau x - L}{x}\right)$ implied by frictional, frictionless optima.

Marginal Terms in Optimal Tax Formula

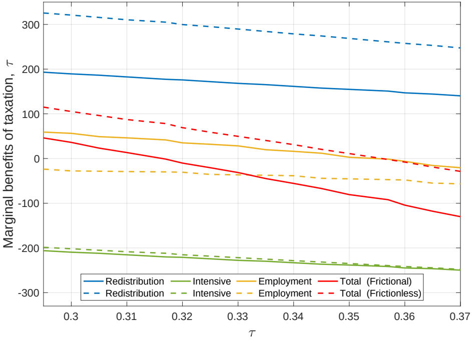


Figure: Marginal terms in frictional and frictionless optimal τ equations.

$$\partial W := -E\left[\left\{\frac{U_c}{\Lambda} - 1\right\}\left\{x + (1-\tau)\frac{\partial q}{\partial \tau}\right\}\right] - \frac{\tau}{1-\tau}E[x\mathcal{E}^c_{x,1-\tau}]$$

$$-\frac{1}{1-\tau}E\left[\left(\left\{\frac{\Delta U}{\Lambda} + \tau x + b - L\right\}\mathcal{E}_{1-u,v} + \left\{\mathcal{E}_{1-u,v} - \frac{vN(q)}{1-u}\right\}q\right)\mathcal{E}_{v,1-\tau}\right]$$

Conclusions

- **Theory:** Incidence of labor taxation falls on firms under frictions. Higher taxes:
  - Modify before-tax income inequality within and across talent markets.
  - Lower vacancy posting, and hence, employment.
- **Theory:** channels may imply higher or lower optimal marginal taxes depending on parameters.
- **Quant analysis:** frictions call for moderately lower (not higher!) taxes.
- **Quant analysis:** mainly due to weakening of redistributive value of taxation

Nonlinear taxation

- General first order condition for arbitrary perturbation of nonlinear taxes:

$$
\begin{aligned}
&-\int_{\mathbb{R}_+} E\left[\frac{M}{\Lambda}\left\{\Omega[x]+(1-T'[x])\partial_\Omega q\right\}+\left\{\frac{T'[x]}{1-T'[x]}\tilde{\eta}-1\right\}(1-T'[x])\partial_\Omega q\,\middle|\, x\right]h(x)dx \\
&\quad=-\int_{\mathbb{R}_+}\{T''[x]\partial_\Omega q\}\frac{T'[x]x}{1-T'[x]}E[\tilde{\mathcal{E}}^c|x]h(x)dx-\int_{\mathbb{R}_+}\partial\mathcal{B}^M(x)\cdot\Omega[x]h(x)dx \\
&\qquad+\int_{\mathbb{R}_+}\frac{1}{1-T'[x]}E\left[\left\{\frac{\Delta U}{\Lambda}+T[x]+b\right\}\mathcal{E}_\mu+(1-T'[x])\partial_\Omega q\,\middle|\, x\right]h(x)dx.
\end{aligned}
$$

Nonlinear taxation

- Holds if for almost all x :

$$0 = -E\left[\frac{M}{\Lambda}\middle| x\right] + \frac{\mathcal{F}(x)}{1-T'[x]} + \partial\mathcal{B}^M(x) + \frac{\mathcal{D}(x)}{1-T'[x]},$$

where

$$\begin{aligned}\mathcal{F}(\mathbf{x}) =& \int_0^1 \frac{\mathbf{M}(\overline{\theta}(\mathbf{x}),\mathbf{i})}{\mathbf{\Lambda}}(\mathbf{1}-\mathbf{T}'[\mathbf{x}(\overline{\theta}(\mathbf{x}),\mathbf{i})])\mathcal{M}(\overline{\theta}(\mathbf{x}),\mathbf{i})\mathbf{di}\cdot\frac{\mathbf{k}(\overline{\theta}(\mathbf{x}))}{\mathbf{h}(\mathbf{x})} \\ &+ \int_0^1 \frac{\mathbf{\Delta U}}{\mathbf{\Lambda}}(\mathbf{x}(\overline{\theta}(\mathbf{x}),\mathbf{i})\mathbf{di}\cdot\frac{\lambda(\overline{\theta}(\mathbf{x}))}{\delta+\lambda(\overline{\theta}(\mathbf{x}))}\frac{\delta}{\delta+\lambda(\overline{\theta}(\mathbf{x}))}\frac{\mathcal{E}_{\lambda,\overline{\mathbf{q}}}(\overline{\theta}(\mathbf{x}))}{\overline{\mathbf{q}}(\overline{\theta}(\mathbf{x}))}\frac{\mathbf{k}(\overline{\theta}(\mathbf{x}))}{\mathbf{h}(\mathbf{x})}\end{aligned}$$

$$\partial\mathcal{B}^M(x) = 1 + \left(\frac{x}{\frac{T'[x]}{1-T'[x]}E[\tilde{\mathcal{E}}^c|x]}\frac{\partial\left(\frac{T'[x]}{1-T'[x]}E[\tilde{\mathcal{E}}^c|x]\right)}{\partial x} + 1 + \frac{xh'(x)}{h(x)}\right)\frac{T'[x]}{1-T'[x]}E[\tilde{\mathcal{E}}^c|x]$$

$$\begin{aligned}\mathcal{D}(\mathbf{x}) =& -\int_0^1 (\mathbf{1}-\mathbf{T}'[\mathbf{x}(\overline{\theta}(\mathbf{x}),\mathbf{i})]\phi(\overline{\theta}(\mathbf{x}),\mathbf{i}))\mathcal{M}(\overline{\theta}(\mathbf{x}),\mathbf{i})\mathbf{di}\cdot\frac{\mathbf{k}(\overline{\theta}(\mathbf{x}))}{\mathbf{h}(\mathbf{x})} \\ &- \int_0^1 \left\{\mathbf{T}[\mathbf{x}(\overline{\theta}(\mathbf{x}),\mathbf{i})] + \mathbf{b} + \chi\mathbf{q}(\overline{\theta}(\mathbf{x}),\mathbf{i})\right\}\mathbf{di}\cdot\frac{\lambda(\overline{\theta}(\mathbf{x}))}{\delta+\lambda(\overline{\theta}(\mathbf{x}))}\frac{\delta}{\delta+\lambda(\overline{\theta}(\mathbf{x}))}\frac{\mathcal{E}_{\lambda,\overline{\mathbf{q}}}(\overline{\theta}(\mathbf{x}))}{\overline{\mathbf{q}}(\overline{\theta}(\mathbf{x}))}\frac{\mathbf{k}(\overline{\theta}(\mathbf{x}))}{\mathbf{h}(\mathbf{x})},\end{aligned}$$

where $\phi := \frac{1}{1+\frac{T''[x]x}{1-T'[x]}\mathcal{E}^c}$.

Taxing with Frictions: Why this Framework?

- Relatively rich frictional model:
  - Job destruction shocks consign workers temporarily to unemployment.
  - Randomly arriving opportunities to climb job ladders and find less extractive firms.
- Burdett-Mortensen element of many structural labor search models.
  - Postel-Vinay-Robin (2002).
- On the job search an ingredient in solving macro search puzzles.
  - Hornstein-Krusell-Violante, (2011).